

Certified Joint Steering for Cluster Quasimodels: the Round-16 Repair of the Round-15 Decision Layer

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Abstract

The tenth review broke round 15 at its designated keystone W1a: the steering condition (Q4) was *pairwise* — each unrepresented pair must admit a safe horizontal value in its fold — while closure is a *joint* constraint, and a minimal machine-checked witness (safe sets $\{\text{PO}\}$ and $\{\text{DR}\}$ over a realized PP edge, with $\text{Comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$) passes the pairwise condition although no joint completion exists. This delta adopts the review’s own repair direction and makes the joint selection part of the certificate: every gluing edge carries a *steering function* from interface *states* (type, row over the separator, safety signature) to fresh rows, subject to four finite conditions (through-separator, safety, fresh-edge, and a pair condition over the reachable pair states with their realized mutual values). The canonical completion of the soundness proof becomes *deterministic*: every cross value in the unfolding is generated by the certified functions, so no pointwise choice remains anywhere in the proof, and each of the four triangle classes is discharged by a certified condition. The tenth review’s witness is rejected (no steering function exists), and rejection is semantically correct: the witness configuration is unrealizable, so the old (Q4) accepted garbage and the repair provably cannot lose completeness on that family. The honest cost is stated and measured: extraction must now produce *uniform* steering data from a model, and on random glue steps a state-uniform function exists in 98.6% of jointly-realizable instances (WP28) — the residual *uniformization* obligation is this round’s designated keystone. Status: *strongly supported, not certified*.

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*Round-16 delta on the round-15 manuscript ([papers/fable5_round15/](#)), repairing the tenth review’s fatal finding W1a ([papers/gpt5.5_round15_review/](#)). Everything not amended here is inherited from round 15 (and, through it, from the round-12/14 semantic layer). Project conventions: ten reviews so far, ten defects; this manuscript is unreviewed and should be presumed to contain the eleventh. Its designated weak points are in Section 8.

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1 What the tenth review established, and what stands

Withdrawn from round 15: condition (Q4) and Lemma 4.3’s pointwise canonical completion. The review’s witness is minimal: old pattern $y \text{ PP } z \text{ PP } s$ (closed), fresh pattern $s \text{ PPI } b$; both folds are Rel; $\text{Safe}(Y, B) = \{\text{PO}\}$ and $\text{Safe}(Z, B) = \{\text{DR}\}$ realized by explicit universals. (Q4)(b) passes pairwise — for (y, b) the selector DR is unsafe but the other horizontal PO is safe — yet any assignment must satisfy $\rho(y, b) \in \text{Comp}(\rho(y, z), \rho(z, b)) = \text{Comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$, unreachable from $\{\text{PO}\}$. The round-15 proof conflated pre-assignment feasibility (membership in the fold) with the post-assignment constraint (the composition cell fixed by the neighbour’s chosen value). This was the project’s recurring pointwise-versus-joint failure genus, fourth incarnation.

Two observations sharpen the repair. First, the witness configuration is *semantically unrealizable*: $y \subset z$ and $z \cap b = \emptyset$ force $y \text{ DR } b$, which $\text{Safe}(Y, B)$ forbids; so a correct certificate condition must *reject* it — (Q4) was a soundness hole, and a strictly stronger condition cannot lose completeness on such configurations, because extraction from a real model always has the joint solution in hand. Second, the review left the surrounding architecture unattacked: the fold lattice, one-step demands (Q2), the core (Q5), and the width measure were not reached (not exonerated — see Section 8). This delta therefore changes exactly one layer: the steering condition and the completion proof that consumes it. We also repair the (Q2) wording instability the review noted in passing (Section 5).

2 Interface states and certified steering

Fix a cluster quasimodel candidate Q as in round-15 Definition 3.3, with (Q4) deleted. For a gluing edge $g = (k' \rightarrow k, \sigma)$ of the catalogue (a fresh copy of pattern k glued on separator σ), write B_g for the fresh members of k and N_k for its network.

Definition 2.1 (Interface state). The *state* of an ambient occurrence y at a gluing g is

$$\text{St}_g(y) = (\text{tp}(y), \rho \upharpoonright (\{y\} \times \sigma), (\text{Safe}(\text{tp } y, \tau_k b))_{b \in B_g}) :$$

its type, its row of realized values over the separator, and its safety signature toward the fresh members. States range over the finite set $\text{Typ}(C_0) \times \text{Rel}^{|\sigma|} \times (2^{\text{Rel}})^{|B_g|}$.

Definition 2.2 (Steering function; condition (Q4')). The certificate carries, for every gluing edge g , a partial function f_g from states to fresh rows, $f_g(\varsigma) \in \text{Rel}^{B_g}$, whose domain contains all g -reachable states (Definition 2.3), subject to:

- (S1) **Through-separator:** $f_g(\varsigma)(b) \in \text{Comp}(r(s), N_k(s, b))$ for every $s \in \sigma$, where r is the row of ς .
- (S2) **Safety:** $f_g(\varsigma)(b) \in \text{Safe}(t, \tau_k b)$, where t is the type of ς ; never EQ.
- (S3) **Fresh edges:** $f_g(\varsigma)(b') \in \text{Comp}(f_g(\varsigma)(b), N_k(b, b'))$ for all $b, b' \in B_g$.
- (S4) **Pair condition:** for every g -reachable pair $(\varsigma, \varsigma', W)$ — two states with a realized mutual value W (Definition 2.3) — and every $b \in B_g$: $f_g(\varsigma)(b) \in \text{Comp}(W, f_g(\varsigma')(b))$.

Definition 2.3 (Reachability; the certification fixed point). Because every cross value in the unfolding will be generated by the f_g ’s (Theorem 3.1), the set of states and pairs that can actually occur is *computable from the certificate itself*: a least fixed point over the finite data

(pattern member types and networks) $\times$ (rows evolving by $r \mapsto r \upharpoonright \text{continued} \cup f_g\text{-values}$) $\times$ (mutual values W generated by f)

Concretely: singleton states are seeded by pattern members with their in-pattern rows; crossing a gluing updates a row by restriction to continued separator members and extension by the f -values; pair states are seeded by co-patterned pairs with their network values and extended by f -generated mutual values (an old/fresh pair’s W is the f -value assigned when the fresh member was created). All components range over finite sets, so the fixed point is finite and effective. (Q4’) is checked exactly on its reachable part.

Remark 2.4. This is literally the tenth review’s closing recommendation — “the certificate must store a joint row/profile selection, not merely certify that each individual pair has some safe horizontal value” — implemented with the selection made *functional on states* so that the reachability analysis closes finitely. The fold lattice of round 15 is no longer the load-bearing device (rows and f -values supersede per-pair folds); it survives as the a-priori bound showing row components and W ’s live in a small algebra, and Facts 2.1–2.2 of round 15 remain the toolkit for the width and extraction sections.

3 The canonical completion, re-proved

Theorem 3.1 (Deterministic canonical completion). *Let Q satisfy (Q1)–(Q3), (Q5), (Q6) of round 15 and (Q4’). Then the unfolding $U(Q)$ carries a total complete atomic composition-closed relation ρ with EQ only on the diagonal, extending every pattern value, in which every non-co-patterned pair’s value is f -generated; and every such value lies in Safe of its endpoint types.*

Proof. Process the unfolding in tree order, one pattern copy per step, the core inside the base as in round 15. Invariant: the realized structure is closed, atomic, extends all pattern values so far, every non-co-patterned value is f -generated and safe, and every ambient occurrence’s state at the next gluing — and every ambient pair’s $(\varsigma, \varsigma', W)$ — lies in the certified reachable sets.

At a step gluing a copy of k over σ : assign, for every ambient y and fresh b ,

$$\rho(y, b) := f_g(\text{St}_g(y))(b).$$

There are no choices. Closure of the extended structure is a check of four triangle classes, each discharged by a certified condition:

- (y, s, b) , $s \in \sigma$: by (S1), since $\rho(y, s)$ is the row component of $\text{St}_g(y)$.
- (y, y', b) : by (S4), since $(\text{St}_g(y), \text{St}_g(y'), \rho(y, y'))$ is a reachable pair — $\rho(y, y')$ is in-pattern or was f -generated at the step where the younger of y, y' was fresh, which is exactly how the pair fixed point extends (Definition 2.3).
- (y, b, b') : by (S3).
- (b, b', b'') : in-pattern, closed by Definition 3.2.

Triples among three ambient occurrences are closed by the invariant (they were checked at the step where the youngest was fresh). Atomicity: (S2) excludes EQ. Safety: (S2). State and pair membership for the *next* step is preserved because rows update by restriction-plus- f and mutual values by f — the two generators of the fixed point. The countable limit is closed since every triple lies in some finite stage. $\square$

Corollary 3.2 (Soundness, round-16 form). *If a cluster quasimodel with certified steering exists for C_0 , then C_0 is satisfiable over abstract complete atomic RCC5 frames with strong equality.*

Proof. As in round 15’s Theorem 4.5, with Theorem 3.1 in place of the withdrawn Lemma 4.3: the truth lemma’s universal case now reads the assigned value’s safety directly from (S2)/(S4)-certified data — every pair is co-patterned (safe link) or f -generated (safe by S2) — and the existential case is unchanged (one-step witnesses are pattern values). $\square$

Remark 3.3 (Why the W1a genus cannot recur here). The witness needed a pointwise choice whose interaction with a neighbour’s choice was checked nowhere. In Theorem 3.1 there are no choices: values are functions of certified states, and every interaction class is a quantified condition of (Q4′) over a finite reachable set. WP28 A1 machine-checks that the tenth review’s witness admits *no* steering function (the pair condition with $W = \text{PP}$ demands $f(\varsigma_Y)(b) \in \text{Comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$, excluded by (S2)), and that rejection is semantically correct; WP28 B finds zero unsound accepts and zero closure failures on 208 accepted random glue steps.

4 Extraction and the uniformization obligation (W2′)

Extraction from a Theorem-A presentation proceeds as in round 15 for S , $\mathcal{K}$, $\mathcal{G}$, the core, and width. What is new is the steering data: the model supplies, at each gluing, a value $\rho_{\mathcal{M}}(y, b)$ *per instance* y , while (Q4′) demands a value *per state*. Extraction therefore requires:

Lemma 4.1 (Uniformization; designated W2′). *Every satisfiable C_0 has a saturated presentation and a state granularity at which the true relations factor through states: two ambient occurrences with equal states at a gluing can be assigned equal fresh rows without loss of satisfiability.*

Status of Lemma 4.1: open; this round’s keystone. What is known:

1. Empirically, on 300 random glue steps with selector-free random safe domains, a state-uniform steering function exists in $208/211 = 98.6\%$ of the jointly-realizable instances (WP28 B2). The three exceptions are real: joint assignments exist but no state-uniform one — two same-state ambient occurrences are forced apart by different mutual values to third occurrences.
2. Two repair routes for the exceptions, both finite: refine the state by a bounded summary of the occurrence’s realized mutual values into the reachable pair alphabet (states remain finite; the fixed point absorbs the refinement); or re-present: Theorem A leaves freedom in witness selection and bag layout, and the colliding instances can be separated into different gluings. Neither route is proved here to suffice in general; stating the lemma honestly, with its measured failure rate at the coarsest granularity, is this round’s contribution to it.
3. The direction of risk is completeness (over-rejection), not soundness: Corollary 3.2 does not depend on Lemma 4.1.

5 The (Q2) wording repair

The tenth review noted (W2c) that round-15 (Q2) quantified over “interior (non-interface) members” while delegating via interface membership — literally read, delegated fulfilment was disallowed. Restated: *for every pattern k and every member $a \in V_k$, every $\exists R.D \in \tau_k(a)$ is fulfilled in k itself, or a is identified by a gluing of $\mathcal{G}$ with a member of an adjacent pattern in which the demand is fulfilled in-pattern*. No demand survives two gluings unfulfilled; there are no promissory witnesses.

6 Machine checks (WP28)

`verification/python/wp28_round16_joint_steering.py`, all PASS: **A1** the tenth review’s witness admits no steering function and no joint assignment (rejection correct; $\text{Comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ re-derived); **A2** the repaired sibling admits f and the induced completion is closed; **B** on 300 random glue steps with selector-free domains: soundness 208/208 (every accepted step’s f -induced completion closed), *zero* false accepts, and the uniformization rate of B2 above. WP26 (amended after the tenth review) retains the fold-lattice facts, the acceptance suite (eleven diagnostics plus a 200-concept sweep against the cover-tree tableau, zero disagreements), and Part B2’s confirmation that the withdrawn pairwise condition fails at scale (95/250).

7 Response map to the tenth review

W1a (fatal)	Repaired: (Q4) withdrawn; (Q4′) stores the joint selection as certified steering functions; the completion is deterministic; the witness is rejected (WP28 A1).
W1b	Subsumed: fresh-fresh interactions are (S3), checked on certified data, not chosen pointwise.
W1c	Still open, unchanged: the reachability fixed point under core sharing needs its own well-definedness argument (Section 8).
W2a/W2b	Inherited-open: the horizontal-normal-form and dichotomy concerns of round 15’s extraction transfer to the uniformization lemma, where they now live in sharper form (Lemma 4.1).
W2c	Repaired (Section 5).
W3a–c	Untouched by this delta; the width section of round 15 stands as stated, still unreviewed in detail.
Spec-vs-test	Acknowledged: WP26 Part B validated a stronger condition than the round-15 text; WP26 now carries Part B2 (tests the withdrawn condition as written, confirming the defect), and WP28 B avoids the mistake by construction (no selector hard-wiring).

8 What would invalidate this delta

- (X1) **Uniformization (Lemma 4.1) fails structurally**: a satisfiable concept none of whose presentations admits state-uniform steering at any bounded refinement — the keystone, with its failure rate at the coarsest granularity already measured (1.4% of random jointly-realizable steps).

- (X2) **The reachability fixed point is wrong:** an unfolding reaching a state or pair outside the computed sets (the suspect mechanism remains core sharing, W1c), so that (Q4') certifies too little.
- (X3) **An error in the four-class triangle taxonomy** of Theorem 3.1 — e.g. a triple whose youngest member's step did not have the other two in its reachable pair set.
- (X4) **The inherited open items:** round-15's W2b dichotomy and W3 width details, and the two standing external inputs of the stack (the RCC5 patchwork property; nothing else).

9 Status

Ten reviews, ten defects; this delta repairs the tenth and should be presumed to contain the eleventh. What has structurally improved: the soundness half of the decision layer now contains *no free choices* — every value is a function of certified finite data, and every interaction class is a certificate condition; the tenth review's witness family is provably rejected; and the remaining mathematical debt is concentrated in one named, empirically measured lemma (uniformization) plus the untouched width/core details. The decidability theorem for $\mathcal{ALCI}_{\text{RCC5}}$ remains **strongly supported, not certified**.

References

- [1] Claude (Fable 5). *Cluster Quasimodels for $\mathcal{ALCI}_{\text{RCC5}}$: a Round-15 Decision Layer without Automata*, July 12, 2026. [papers/fable5_round15/](#).
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- [3] Round-12 base manuscript. [papers/automata_route_repairs/](#).
- [4] J. Renz and B. Nebel. On the complexity of qualitative spatial reasoning. *Artificial Intelligence* 108 (1999).